

TD 1 : LINUX GIT PYTHON

Question 5:

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Problèmes  Sortie  Console de débogage  Terminal  Ports

●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ python --version
Python 3.14.2
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ cat data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000

...

Problèmes  Sortie  Console de débogage  Terminal  Ports
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●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ cat data/sample/prices.csv
2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
2026-09-29,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ head -3 data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ pwd
bash: pwd: command not found
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ pwd
/workspaces/esilv-marketpulse-g07-Tryharders
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ ls
ls -la
CONTRIBUTING.md  README.md  TEAM.md  TEAM_TEMPLATE.md  config  data  evidence  lab  readiness  requirements.txt  src
total 68
drwxr-xr-x 9 codespace root 4096 Oct 7 16:46 .
drwxr-xr-x 5 codespace root 4096 Oct 7 16:46 ..
drwxr-xr-x 8 codespace root 4096 Oct 7 16:46 .git
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ ls -la
total 68
drwxr-xr-x 9 codespace root 4096 Oct 7 16:46 .
drwxr-xr-x 5 codespace root 4096 Oct 7 16:46 ..
drwxr-xr-x 8 codespace root 4096 Oct 7 16:46 .git
-rw-rw-rw- 1 codespace root 178 Oct 7 16:46 .gitignore
-rw-rw-rw- 1 codespace root 2943 Oct 7 16:46 CONTRIBUTING.md
-rw-rw-rw- 1 codespace root 10270 Oct 7 16:46 README.md
-rw-rw-rw- 1 codespace root 1474 Oct 7 16:46 TEAM.md
-rw-rw-rw- 1 codespace root 1446 Oct 7 16:46 TEAM_TEMPLATE.md
drwxr-xr-x 2 codespace root 4096 Oct 7 16:46 config
drwxr-xr-x 3 codespace root 4096 Oct 7 16:46 data
drwxr-xr-x 2 codespace root 4096 Oct 7 16:46 evidence
drwxr-xr-x 2 codespace root 4096 Oct 7 16:46 lab
drwxr-xr-x 2 codespace root 4096 Oct 7 16:46 readiness
-rw-rw-rw- 1 codespace root 133 Oct 7 16:46 requirements.txt
drwxr-xr-x 2 codespace root 4096 Oct 7 16:46 src
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ ls -la
total 68
drwxr-xr-x 9 codespace root 4096 Oct 7 16:46 .
drwxr-xr-x 5 codespace root 4096 Oct 7 16:46 ..
drwxr-xr-x 8 codespace root 4096 Oct 7 16:46 .git
-rw-rw-rw- 1 codespace root 178 Oct 7 16:46 .gitignore
-rw-rw-rw- 1 codespace root 2943 Oct 7 16:46 CONTRIBUTING.md
-rw-rw-rw- 1 codespace root 10270 Oct 7 16:46 README.md
-rw-rw-rw- 1 codespace root 1474 Oct 7 16:46 TEAM.md
-rw-rw-rw- 1 codespace root 1446 Oct 7 16:46 TEAM_TEMPLATE.md
drwxr-xr-x 2 codespace root 4096 Oct 7 16:46 config
drwxr-xr-x 3 codespace root 4096 Oct 7 16:46 data
drwxr-xr-x 2 codespace root 4096 Oct 7 16:46 evidence
drwxr-xr-x 2 codespace root 4096 Oct 7 16:46 lab
drwxr-xr-x 2 codespace root 4096 Oct 7 16:46 readiness
-rw-rw-rw- 1 codespace root 133 Oct 7 16:46 requirements.txt
drwxr-xr-x 2 codespace root 4096 Oct 7 16:46 src
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ cd data
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders/data (esther) $ pwd
/workspaces/esilv-marketpulse-g07-Tryharders/data
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders/data (esther) $ ls
sample
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders/data (esther) $ cd sample
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders/data/sample (esther) $ pwd
/workspaces/esilv-marketpulse-g07-Tryharders/data/sample
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders/data/sample (esther) $ ls
bloomberg_reference_expected.json  bloomberg_reference_sample.json  instruments.json  prices.csv
●@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders/data/sample (esther) $

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Question 6 :

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@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ cd ../../
@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ pwd
bash: pwd: command not found
@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ pwd
/workspaces/esilv-marketpulse-g07-Tryharders
@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ █

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@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ █

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@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ grep AAPL data/sample/prices.csv
2026-09-14,AAPL,256.60,258.30,255.40,256.00,42500000
2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000
2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000
2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
2026-09-18,AAPL,259.30,261.00,258.10,260.10,41000000
2026-09-21,AAPL,259.10,261.10,257.90,259.00,42500000
2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000
2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000
2026-09-24,AAPL,261.00,265.00,261.00,263.50,39500000
2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000
2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ grep SP500 data/sample/prices.csv
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,SP500,6623.00,6643.00,6611.00,6628.00,0
2026-09-09,SP500,6640.00,6657.00,6628.00,6642.00,0
2026-09-10,SP500,6642.00,6665.00,6630.00,6650.00,0
2026-09-11,SP500,6642.00,6662.00,6630.00,6647.00,0
2026-09-14,SP500,6663.00,6680.00,6651.00,6665.00,0
2026-09-15,SP500,6664.00,6687.00,6652.00,6672.00,0
2026-09-16,SP500,6663.00,6683.00,6651.00,6668.00,0
2026-09-17,SP500,6681.00,6700.00,6671.00,6685.00,0
2026-09-18,SP500,6684.00,6707.00,6672.00,6692.00,0
2026-09-21,SP500,6683.00,6703.00,6671.00,6688.00,0
2026-09-22,SP500,6701.00,6720.00,6691.00,6705.00,0
2026-09-23,SP500,6704.00,6727.00,6692.00,6712.00,0
2026-09-24,SP500,6715.00,6735.00,6703.00,6720.00,0
2026-09-25,SP500,6711.00,6730.00,6701.00,6715.00,0
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ █

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Question

7:

- @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) \$ python --version
Python 3.14.2
- @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) \$ git --version
git version 2.55.0
- @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) \$ █

```

@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ █

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- [] I can use pwd : afficher le dossier où je suis
- [] I can use ls : Lister le contenu d'un dossier
- [] I can use cd : change directory : tu entres dans le dossier
- [] I can use cat: lis le contenu
- [] I can use head: affiche les 10 premières lignes
- [] I can use grep: affiche seulement lignes contenant un mot.

OPTIONAL:

```
● @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ grep -c AAPL data/sample/prices.csv
21
● @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ grep -c SP500 data/sample/prices.csv
21
○ @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ █
```

Les deux séries ont les mêmes dates, donc on peut comparer Apple et le S&P 500 jour par jour, sur la même période.

```
● @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ ls -lh
total 52K
-rw-rw-rw- 1 codespace root 2.9K Oct 7 16:46 CONTRIBUTING.md
-rw-rw-rw- 1 codespace root 11K Oct 7 16:46 README.md
-rw-rw-rw- 1 codespace root 1.5K Oct 7 16:46 TEAM.md
-rw-rw-rw- 1 codespace root 1.5K Oct 7 16:46 TEAM_TEMPLATE.md
drwxrwxrwx+ 2 codespace root 4.0K Oct 7 16:46 config
drwxrwxrwx+ 3 codespace root 4.0K Oct 7 16:46 data
drwxrwxrwx+ 2 codespace root 4.0K Oct 7 16:46 evidence
drwxrwxrwx+ 2 codespace root 4.0K Oct 7 16:46 labs
drwxrwxrwx+ 2 codespace root 4.0K Oct 7 16:46 readlines
-rw-rw-rw- 1 codespace root 133 Oct 7 16:46 requirements.txt
drwxrwxrwx+ 2 codespace root 4.0K Oct 7 16:46 src
● @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ head -n 5 data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.00,252.00,249.00,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
● @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ tail -n 5 data/sample/prices.csv
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
○ @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ █
```

- ls shows only names.
- ls -la adds details and hidden files.
- ls -lh shows details with readable sizes.
- head shows the first lines of a file
- tail the last lines.

```
● @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ git status
On branch esther
Your branch is up to date with 'origin/esther'.

nothing to commit, working tree clean
● @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ git log --oneline
981664e (HEAD -> esther, origin/esther) Ajout du fichier TEAM.md
5207db8 (origin/main, origin/leopold, origin/jeanne, origin/branchemaelys, origin/HEAD) first td session
ba5c9f4 Update Readme
9fd6c8a Update Readme
bfb94a6 init projecttemplate
○ @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ █
```

TD 2 :

load_instruments() : ouvre instruments.json et le lit avec json.load

load_prices() : ouvre prices.csv et le lit avec csv.DictReader

main() : appelle les autres fonctions dans l'ordre et affiche le résultat

```

@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ python -c "import json; d = json.load(open('data/sample/instruments.json')); print(d['instrument']); print(d['benchmark']['ticker'])"
{'ticker': 'AAPL', 'name': 'Apple Inc.', 'currency': 'USD', 'market': 'NASDAQ'}
SP500
@ecetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ █

```

0 L 31, col 5 Espaces : 4 UTF-8

Codespaces: effective festa | esther* | 0 | 0 | 0 | L 37, col 5 | Espaces : 4 | UTF-8 | LF | Python | Se connecter | Layout: French

```
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
@getetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ python src/main.py | head -5
class 'list'
class 'dict'
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}
=== MarketPulse ===

@getetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ █

--- marketpulse ---
@getetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ python src/main.py | head -5
250.00
<class 'str'>
--- MarketPulse ---

Instrument
@getetere →/workspaces/esilv-marketpulse-g07-Tryharders (esther) $ █

1/0 0 L 33, col 5 Expires: 4 UTF-8 LF {}
```

```

@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ python src/main.py | head -5
250.00
<class 'str'>
<class 'float'>
=== MarketPulse ===

@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ █

```

```

==== MarketPulse ====

@ecetere →/workspaces/esilv-marketpulse-g07-Tryharden (esther) $ python src/main.py | head -5
250.00
<class 'str'>
<class 'float'>
21
21
@ecetere →/workspaces/esilv-marketpulse-g07-Tryharden (esther) $

```

```
21
● @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ python src/main.py | head -5
250.0
6600.0
266.2
6742.0
=== MarketPulse ===
```

```
def display_market_summary(asset, prices, show_currency=True):
    currency = ""
    if show_currency:
        currency = " " + asset["currency"]

    print(f"{asset['ticker']} - {asset['name']}")
    print(f"Observations : {len(prices)}")
    print(f"First close : {get_first_close(prices):.2f}{currency}")
    print(f>Last close : {get_last_close(prices):.2f}{currency}")
```

```
instruments = load_instruments()
```

```
@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ python src/main.py
AAPL - Apple Inc.
Observations : 21
First close  : 250.00 USD
Last close   : 266.20 USD

SP500 - S&P 500
Observations : 21
First close  : 6600.00
Last close   : 6742.00

@ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $
```

Question 8 :

```
● @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ python src/main.py
=== MarketPulse ===

Market configuration
Period : 1 month
Interval : Daily

Instrument
AAPL - Apple Inc.
Observations : 21
First close : 250.00 USD
Last close : 266.20 USD

Benchmark
SP500 - S&P 500
Observations : 21
First close : 6600.00
Last close : 6742.00
○ @ecetere → /workspaces/esilv-marketpulse-g07-Tryharders (esther) $ █
```